NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Trauma

1. hip and femoral fractures
2. pelvis and acetabulum
3. tibial plateau and pilon
4. ankle fractures
5. proximal humerus and shoulder girdle
6. distal radius and carpus
7. elbow and forearm
8. fracture healing and fixation principles
9. polytrauma, open fractures and compartment syndrome

Hip and femoral fractures

1. Applied anatomy of the proximal femur

- **Neck-shaft angle** about 125-135 degrees; **anteversion** about 10-15 degrees in adults.
- **Blood supply of the femoral head:** mainly the **lateral epiphyseal (retinacular) branches of the medial femoral circumflex artery (MFCA)**, running up the posterosuperior neck. Lesser contributions: inferior gluteal, LFCA (anterior neck), and the artery of the ligamentum teres (small in adults; important in children). **Intracapsular fractures** disrupt the retinacular vessels: **AVN and non-union**.
- **Calcar femorale:** dense posteromedial cortical bone (load-bearing); **Ward's triangle** is the weak area between trabecular groups (Singh index grades osteoporosis from trabecular patterns).
- The **intertrochanteric line and capsule:** the capsule attaches anteriorly to the intertrochanteric line, but posteriorly only halfway down the neck.

2. Femoral neck fractures

Classifications

Garden	Description	Practical grouping
I	Incomplete / **valgus impacted**	Undisplaced
II	Complete, undisplaced	Undisplaced
III	Complete, **partially displaced** (trabeculae misaligned)	Displaced
IV	Complete, **fully displaced** (trabeculae of head realign with the acetabulum)	Displaced

- **Pauwels** (angle of the fracture line from horizontal; for young, high-energy fractures): **I under 30 degrees**, **II 30-50 degrees**, **III over 50 degrees (vertical shear, highest non-union and fixation failure)**.
- Location: subcapital, transcervical, basicervical (behaves like an extracapsular fracture).

Elderly (low-energy)

- **Undisplaced (Garden I-II): cannulated screws** (3 parallel screws in an inverted triangle, the inferior screw on the calcar, posterior screw against the posterior cortex; no screw entry below the lesser trochanter level: subtrochanteric fracture risk) or a sliding hip screw. Some evidence favours arthroplasty in very elderly frail patients with posterior tilt over about 20 degrees (high failure).
- **Displaced (Garden III-IV): arthroplasty**:
  - **Total hip arthroplasty** for active, independent, cognitively intact patients (better function; **HEALTH trial** showed small functional gains with more dislocations; NICE recommends THA for those who walk independently outdoors with a stick or less, are not cognitively impaired and are medically fit).
  - **Hemiarthroplasty** (bipolar or unipolar) for frailer, less active patients. **Cemented stems** are recommended (less pain, fewer periprosthetic fractures, better function: **WHiTE 5**), with care for **bone cement implantation syndrome**.
- **Orthogeriatric care:** surgery **within 36-48 hours** (NICE: on the day of or day after admission), early mobilisation, delirium prevention, nutrition, bone health (**secondary fracture prevention**: bisphosphonate, vitamin D), falls assessment. **One-year mortality about 25-30%.**

Young (high-energy)

- An **emergency**: **urgent anatomical reduction** (closed or open, Watson-Jones or Smith-Petersen approach) and **stable fixation** (cannulated screws, sliding hip screw ± derotation screw, or a fixed-angle device such as a **dynamic hip screw with anti-rotation screw** or **proximal femoral locking/medial buttress plate** for vertical Pauwels III fractures).
- Complications: **AVN (about 20-30%)**, **non-union (about 10-30%)** (valgus intertrochanteric osteotomy (**Pauwels osteotomy**) converts shear into compression for non-union in young patients).
- **Ipsilateral femoral neck and shaft fracture:** the neck fracture (vertical, basicervical, undisplaced) is **missed in about 20-30%**: get a dedicated CT/fine-cut through the neck. Prioritise the **neck** (anatomical reduction).

Stress fractures of the femoral neck (Fullerton/Devas)

- **Compression (inferomedial)**: protected weight bearing; fix if the line exceeds about 50% of the neck or does not heal.
- **Tension (superolateral)**: high risk of displacement: **fix with screws**. Female athlete triad / RED-S.

3. Intertrochanteric (pertrochanteric) fractures

- Extracapsular; good blood supply, **unions reliably**; problem is fixation failure (**cut-out**).
- **Classification (AO/OTA 31A):**
  - **A1:** simple pertrochanteric (two-part; stable; intact lateral wall).
  - **A2:** multifragmentary pertrochanteric (posteromedial comminution, lateral wall incompetent in A2.2-2.3 per 2018).
  - **A3: intertrochanteric/reverse oblique** (unstable).
  - Evans (stable vs unstable; reverse obliquity is unstable), Boyd and Griffin, Jensen.
- **Unstable features: posteromedial comminution** (lesser trochanter), **reverse obliquity**, **subtrochanteric extension**, **lateral wall fracture** (thickness under about 20.5 mm predicts failure; Hsu), and inability to reduce.
- **Implants:**
  - **Sliding (dynamic) hip screw (DHS)** for **stable** fractures (A1): controlled collapse along the barrel.
  - **Cephalomedullary nail** (PFN, PFNA, Gamma, TFN, InterTAN) for **unstable** fractures, **reverse oblique** and subtrochanteric extension, deficient lateral wall. **DHS is contraindicated in reverse obliquity** (the fracture slides medially along the lag screw axis).
- **Tip-apex distance (TAD, Baumgaertner):** sum of the distances from the lag screw tip to the head apex on AP and lateral views, corrected for magnification. **Keep under 25 mm** (cut-out risk rises sharply above 25 mm). Position **centre-centre or inferior-centre** on AP, centre on lateral.
- **Complications: cut-out** (commonest mechanical failure; poor TAD, superior/anterior screw placement, unstable pattern), **Z-effect** (two-screw nails), **varus collapse**, periprosthetic fracture at the tip of a short nail, non-union (rare), malrotation.

4. Subtrochanteric fractures

- Within **5 cm distal to the lesser trochanter**. **High stress concentration** (medial compression, lateral tension).
- Deforming forces on the proximal fragment: **flexion (iliopsoas), abduction (gluteus medius/minimus) and external rotation (short external rotators)**; the shaft is pulled into **adduction (adductors)** and shortening.
- **Russell-Taylor** (piriformis fossa and lesser trochanter involvement); Seinsheimer.
- **Treatment: cephalomedullary (reconstruction) nail**, often after **open or clamp-assisted reduction** (reduction is the key; a malreduced varus fracture fails). Plates (95-degree blade plate, locking plates) in selected cases.
- **Atypical femoral fractures (AFF) with long-term bisphosphonates (ASBMR criteria):** subtrochanteric or diaphyseal, **minimal or no trauma**, **transverse or short oblique** with a **medial spike** when complete, **non-comminuted**, **lateral cortical "beaking"/thickening**, prodromal thigh pain, often **bilateral** (image the other femur). **Stop the bisphosphonate**, calcium/vitamin D, consider **teriparatide**; **prophylactic intramedullary nailing** for incomplete fractures with pain or a visible fracture line.

5. Femoral shaft fractures

- High-energy injuries; blood loss up to about 1-1.5 litres; **fat embolism** risk; associated injuries (ipsilateral neck fracture, knee ligament injuries, hip dislocation).
- **Winquist-Hansen** comminution grades 0-IV.
- **Treatment of choice: reamed, statically locked antegrade intramedullary nail** within 24 hours in a stable patient. **Early fixation reduces pulmonary complications** (ARDS, fat embolism) and allows mobilisation.
- **Antegrade nail entry: piriformis fossa** (straight nails) or **trochanteric** entry (curved-proximal nails; avoids iatrogenic neck damage; small abductor damage). **Retrograde nail** (through the intercondylar notch): obesity, bilateral fractures, **ipsilateral femoral neck fracture** (with separate neck fixation), floating knee, pregnancy, distal fractures, ipsilateral acetabular fracture.
- **Reaming** increases union rates (**SPRINT for tibia** showed no harm; for femur, reamed nails union better); theoretical fat embolism risk.

- **Damage control orthopaedics (DCO): external fixation first**, nail later (within about 2 weeks; ideally day 4-14 when the inflammatory "second hit" settles), in **borderline or unstable** polytrauma patients (see polytrauma notes).
- **Complications:**
  - **Malrotation** (commonest malalignment after nailing; compare with the other side; over about 15 degrees is significant).
  - **Limb-length discrepancy**, **non-union** (about 1-5%; exchange nailing for aseptic hypertrophic non-union), malunion (valgus with proximal fractures, apex-anterior with distal entry).
  - **Fat embolism syndrome:** 24-72 hours, **Gurd and Wilson criteria** (major: respiratory insufficiency, cerebral signs, **petechial rash** (axilla, conjunctiva)); supportive care.
  - Pudendal nerve palsy (traction table perineal post), heterotopic ossification at the entry, knee pain (retrograde).

6. Distal femur fractures

- **AO/OTA 33A (extra-articular), B (partial articular; B3 = coronal "Hoffa" fracture), C (complete articular).**
- **Hoffa fracture:** coronal plane fracture of a femoral condyle (**lateral more common**); seen only on lateral views/CT; fix with **anteroposterior lag screws** (perpendicular to the fracture).
- **Treatment: lateral locking plate** (MIPO) or **retrograde nail** (extra-articular and simple articular); **dual plating (lateral + medial)** for comminuted, bone loss or osteoporotic fractures and in nonunions; **distal femoral replacement** in elderly osteoporotic patients with very distal or periprosthetic fractures.
- **Deforming forces:** the **gastrocnemius** pulls the distal fragment into **extension (apex posterior)**; quadriceps and hamstrings shorten.
- **Periprosthetic distal femur fracture above a TKA:** Lewis-Rorabeck (I undisplaced, II displaced with a stable prosthesis, III loose prosthesis) or Su classification; fix (plate, retrograde nail if the box allows) or revise/replace if loose.

7. Patella (often with knee trauma)

- Undisplaced with intact extensor mechanism (can straight-leg raise): cylinder cast/brace.
- **Displaced (step-off over 2 mm or gap over 3 mm) or extensor lag: tension-band wiring** (K-wires + figure-of-eight; converts tension to compression) or cannulated screws with tension band; partial patellectomy for comminuted inferior pole.
- **Bipartite patella:** superolateral (Saupe type III), bilateral, smooth rounded margins: not a fracture.

> Exam pearl: **Garden III-IV in the elderly = arthroplasty (cemented; THA if active and cognitively intact). Young displaced neck fracture = emergency anatomical reduction. Pauwels III = vertical shear, highest failure.**

> Exam pearl: **TAD under 25 mm prevents cut-out. DHS for stable; nail for unstable, reverse oblique and subtrochanteric. Atypical femoral fracture: lateral cortical beaking, transverse, bilateral: stop bisphosphonates, consider teriparatide, nail.**

> Exam pearl: **Ipsilateral neck + shaft fracture: the neck is often missed; prioritise the neck.**

Pelvis and acetabulum

1. Applied anatomy

- The pelvic ring is stable because of its **posterior ligamentous complex**: the **posterior sacroiliac ligaments (strongest ligaments in the body)**, interosseous SI ligaments, sacrospinous (resists external rotation) and sacrotuberous (resists vertical shear and flexion) ligaments, and the iliolumbar ligaments. **Posterior stability matters more than anterior.**
- Anterior: pubic symphysis (fibrocartilage); normal symphyseal width under about 5 mm, SI joint under about 4 mm.
- **Bleeding:** about 80-90% is **venous** (presacral and prevesical plexus) or from cancellous bone; about 10-15% is **arterial**, most commonly the **superior gluteal artery** (posterior, at the greater sciatic notch), then internal pudendal and obturator. The **corona mortis** (anastomosis between external iliac/inferior epigastric and obturator vessels, behind the superior pubic ramus, about 4-6 cm from the symphysis) is at risk in the Stoppa and ilioinguinal approaches.
- **Nerves:** the **L5 root** crosses the sacral ala (at risk with iliosacral screws placed too anteriorly); sacral roots in zones II/III sacral fractures; the lumbosacral trunk.

2. Pelvic ring injuries

Radiographs and CT

- AP pelvis, **inlet view** (beam angled caudally about 25-45 degrees: shows **AP displacement and rotation**) and **outlet view** (beam angled cephalad about 40-45 degrees: shows **vertical displacement** and sacral foramina). **CT** is standard.
- **L5 transverse process avulsion** suggests vertical instability. **Ischial spine avulsion** suggests sacrospinous disruption.

Young-Burgess (mechanism-based; predicts resuscitation needs)

Type	Mechanism	Injury pattern	Notes
APC I	Anteroposterior compression	Symphysis widening **under 2.5 cm**, SI ligaments intact	Stable; non-operative

Type	Mechanism	Injury pattern	Notes
APC II	AP compression	Symphysis **over 2.5 cm**, **anterior SI ligaments torn**, posterior SI intact ("open book")	Rotationally unstable, vertically stable
APC III	AP compression	Anterior and **posterior SI ligaments torn** (complete SI disruption)	Rotationally and vertically unstable; **highest blood loss and transfusion needs**
LC I	Lateral compression	Sacral buckle/impaction + oblique (horizontal) rami fractures	Stable
LC II	Lateral compression	Plus **iliac wing (crescent) fracture** or posterior SI disruption	Rotationally unstable
LC III	Lateral compression	**"Windswept pelvis"**: LC on one side, APC (open book) on the other	Unstable
VS	Vertical shear	Vertical displacement of a hemipelvis through SI joint, sacrum or ilium	Completely unstable; ipsilateral L5 transverse process avulsion
CM	Combined mechanism	Mixed patterns	

- **Lateral compression is the commonest pattern** (often elderly, side impact); **APC is associated with more bleeding**; **LC with head and chest injuries**.

Tile (stability-based)

- **A:** stable (avulsions, iliac wing, transverse sacrococcygeal).
- **B:** rotationally unstable, vertically stable (B1 open book, B2 ipsilateral LC, B3 contralateral/bucket-handle).
- **C:** rotationally **and vertically unstable** (C1 unilateral, C2 bilateral, C3 with acetabular fracture).

Sacral fractures (Denis zones)

Zone	Location	Neurological injury
I	Lateral to the foramina (ala)	About 6%; **L5 root**
II	Through the foramina	About 28%; sciatica (unilateral roots)
III	Medial to the foramina (central canal)	About 57%; **bowel, bladder and sexual dysfunction**; includes **U/H-type (spinopelvic dissociation)** fractures

Emergency management of the bleeding pelvis

1. **ATLS**, two large-bore lines, **massive transfusion protocol** (balanced 1:1:1), **tranexamic acid within 3 hours** (CRASH-2).
2. **Pelvic binder at the level of the greater trochanters** (not the iliac crests) to reduce volume in APC injuries; internally rotate and tie the legs. Avoid over-compressing LC injuries.
3. **FAST** to look for intraperitoneal bleeding.
4. **Haemodynamically unstable despite resuscitation: preperitoneal pelvic packing** (with external fixation) and/or **angioembolisation** (arterial blush on CT; embolisation is the choice for arterial bleeding); **REBOA** in selected centres. Laparotomy if FAST positive.
5. **Anterior external fixation** (iliac crest or **supra-acetabular pins**; supra-acetabular gives better control, risk to the **lateral femoral cutaneous nerve**) or a **C-clamp** for posterior instability (risk: penetration through the greater sciatic notch).

Definitive fixation

- **Symphysis: Pfannenstiel** incision, plating of the superior surface (open book APC II/III).
- **Posterior ring: iliosacral screws** (percutaneous; S1 and S2 corridors; need good inlet/outlet/lateral imaging; danger to **L5 root** anteriorly and **S1 root** through the foramen; **sacral dysmorphism** narrows the S1 corridor), anterior SI plating, posterior tension-band plating, **lumbopelvic fixation** for spinopelvic dissociation.
- **Fragility fractures of the pelvis (FFP, Rommens and Hofmann):** elderly, low-energy LC-type injuries with sacral involvement; mobilise; fix

(percutaneous screws) if pain prevents mobilisation or the fracture progresses.

Associated injuries and complications

- **Urethral injury** (males, mostly **posterior (membranous-bulbar) urethra**, APC and straddle injuries): **blood at the meatus, high-riding prostate, perineal haematoma**: **retrograde urethrogram before a catheter**; suprapubic catheter.
- **Bladder rupture** (extraperitoneal commonest; repair intraperitoneal rupture surgically).
- **Open pelvic fractures** (perineal or rectal wounds): mortality up to about 50%; **diverting colostomy** for rectal/perineal wounds.
- **Morel-Lavallee lesion:** closed degloving over the greater trochanter (shearing off the fascia); infected easily; drain/debride (percutaneous or open) before or with fixation.
- **DVT/PE** (high risk; chemoprophylaxis), **neurological deficit** (L5, S1), sexual dysfunction, chronic SI pain, malunion with limb-length discrepancy and sitting imbalance.

3. Acetabular fractures

Columns and radiology

- **Anterior column:** iliac crest (anterior part) to the pubic symphysis, including the anterior wall. **Posterior column:** greater sciatic notch to the ischial tuberosity, including the posterior wall.
- **Judet views** (45-degree obliques): **obturator oblique** (shows the **anterior column** and **posterior wall**) and **iliac oblique** (shows the **posterior column** and **anterior wall**). Six radiographic lines on AP: **iliopectineal line (anterior column)**, **ilioischial line (posterior column)**, teardrop, roof, anterior rim, posterior rim.
- **Spur sign** (obturator oblique): the intact iliac bone above the joint projecting, seen in **both-column fractures** (no articular segment attached to the axial skeleton).
- **Roof arc angles (Matta):** medial, anterior and posterior roof arcs **over 45 degrees** mean the fracture spares the weight-bearing dome (non-operative may be possible).

Letournel classification

Elementary (5)	Associated (5)
Posterior wall (**commonest overall**)	**Both column** (commonest associated; no articular fragment attached to the ilium; spur sign)
Posterior column	T-type
Anterior wall	Transverse + posterior wall (common)
Anterior column	Posterior column + posterior wall
Transverse (only elementary type through both columns)	Anterior column/wall + posterior hemitransverse

- **Elderly:** anterior column / anterior wall patterns with **quadrilateral plate** medialisation and **superomedial dome impaction ("gull sign")** are typical (low-energy falls).

Management

- **Hip dislocation with acetabular fracture: urgent reduction** (within about 6 hours to reduce AVN), check the sciatic nerve, skeletal traction if unstable.
- **Non-operative:** undisplaced (under about 2 mm), roof arcs over 45 degrees with a congruent joint out of traction, secondary congruence of both-column fractures, small posterior wall fragments with a stable hip (**stress EUA**), elderly patients unfit for surgery.
- **Operative indications: articular displacement over 2 mm** in the dome, **unstable hip** (posterior wall over about 40-50% of the wall is usually unstable; under 20% usually stable), **incarcerated intra-articular fragments**, marginal impaction. Fix within about 2 weeks (the window closes by about 3 weeks).
- **Approaches:**

Approach	Access	Structures at risk
Kocher-Langenbeck (posterior; patient prone or lateral)	Posterior wall and column, transverse, T	**Sciatic nerve** (keep hip extended, knee flexed), **superior gluteal vessels/nerve** at the notch, **medial femoral circumflex artery** (preserve quadratus femoris and obturator externus)

Approach	Access	Structures at risk
Ilioinguinal (Letournel; 3 windows)	Anterior column/wall, both column	**Femoral nerve and vessels**, **lateral femoral cutaneous nerve**, **corona mortis**, spermatic cord, inguinal hernia
Modified Stoppa (anterior intrapelvic)	Quadrilateral plate, anterior column, both column	**Corona mortis**, obturator nerve and vessels
Extended iliofemoral	Both columns (complex late cases)	Highest **heterotopic ossification** risk

- **Elderly:** fix-and-replace (acute THA with acetabular fixation) for severe dome impaction, femoral head damage or pre-existing arthritis.

Complications

- **Post-traumatic arthritis** (commonest late complication; related to the accuracy of reduction: anatomical reduction under 1 mm gives the best results).
- **Heterotopic ossification** (posterior and extensile approaches; prophylaxis with **indomethacin** or **single-dose radiotherapy**; Brooker classification).
- **AVN of the femoral head** (with posterior dislocation), **sciatic nerve palsy** (peroneal division most affected; about 10-15%, partly iatrogenic), **DVT/PE**, infection, malreduction.

4. Hip dislocation (often with acetabular injury)

- **Posterior (about 90%):** dashboard injury, hip **flexed, adducted and internally rotated**; associated posterior wall fracture, **femoral head fracture**, **sciatic nerve injury** (peroneal division), and **ipsilateral knee injury (PCL, patella)**.
- **Anterior:** flexed, abducted and externally rotated; femoral nerve and vessels at risk.
- **Reduce urgently (within 6 hours)**; then CT for intra-articular fragments and head/wall fractures. **AVN** risk rises with delayed reduction (about 5-10% overall).
- **Thompson-Epstein** (posterior dislocation types I-V) and **Pipkin** (femoral head fractures with posterior dislocation):

Pipkin	Description	Treatment
I	Head fracture **below (caudal to) the fovea** (non-weight-bearing)	Non-operative if reduced, or excise small fragments
II	Head fracture **above (cephalad to) the fovea** (weight-bearing)	ORIF (often anterior Smith-Petersen or surgical hip dislocation)
III	I or II + **femoral neck fracture**	ORIF of neck (young) or arthroplasty (older); worst prognosis (AVN)
IV	I or II + **acetabular fracture**	Fix the acetabulum and head

Exam pearl: **APC III = most bleeding; LC = commonest; superior gluteal artery = commonest arterial bleeder; binder at the greater trochanters.** Retrograde urethrogram before catheterising a male with blood at the meatus.

Exam pearl: **Obturator oblique shows anterior column + posterior wall; iliac oblique shows posterior column + anterior wall. Spur sign = both-column fracture.** Posterior wall = commonest acetabular fracture; Kocher-Langenbeck endangers the sciatic nerve and MFCA.

Tibial plateau and pilon

1. Tibial plateau fractures

Anatomy and mechanism

- The **lateral plateau** is **convex** and higher; the **medial plateau** is **concave**, larger and stronger (more load).
- Mechanism: **valgus load with axial force** (lateral plateau, commonest), varus (medial plateau, higher energy, often **fracture-dislocation**).
- **Lateral plateau fractures are commonest (about 55-70%).**
- Associated injuries: **lateral meniscus tears** (commonest with lateral plateau; often peripheral and incarcerated), **medial meniscus**, **MCL and ACL injuries**, **peroneal nerve** (lateral fracture-dislocation), **popliteal artery** (medial plateau and Schatzker IV/VI, bicondylar fractures; **ankle-brachial index under 0.9 = arterial study**), **compartment syndrome** (especially Schatzker VI).

Schatzker classification

Type	Description	Typical patient / mechanism	Treatment principle
I	Lateral **split** (wedge)	Young, dense bone	Percutaneous screws or buttress plate; check for meniscal entrapment
II	Lateral **split-depression**	**Commonest**	Elevate the depressed fragment, **subchondral bone graft/substitute**, raft screws, lateral buttress plate
III	Lateral **pure depression**	**Osteoporotic** elderly	Elevation through a cortical window, graft, raft screws (IIIa lateral, IIIb central)
IV	**Medial plateau**	High energy; fracture-dislocation (Moore); **popliteal artery and peroneal nerve at risk**	Posteromedial approach, buttress plate; poor prognosis

Type	Description	Typical patient / mechanism	Treatment principle
V	**Bicondylar** with metaphysis intact	High energy	Dual plating / lateral locking ± medial plate
VI	Bicondylar with **metaphyseal-diaphyseal dissociation**	Highest energy; **compartment syndrome**	**Staged**: spanning external fixator, then ORIF (dual incisions) when soft tissues allow; or hybrid/circular frame

- **Three-column concept (Luo; CT-based):** lateral, medial and **posterior** columns (the posterior column, especially **posteromedial fragments**, is missed by Schatzker; needs a **posteromedial (Lobenhoffer) or posterior approach** and buttress plating).
- **Moore classification** (fracture-dislocations): split, entire condyle, rim avulsion, rim compression, four-part.

Investigation

- AP, lateral, oblique; **CT** for every operative plateau; **MRI** shows meniscal/ligament injuries (commonly present).
- **Assess soft tissues** (fracture blisters, swelling: **"wrinkle sign"** shows readiness for surgery, usually 7-14 days).

Management

- **Non-operative:** undisplaced or minimally displaced, stable to varus/valgus stress in full extension (under about 10 degrees), elderly low-demand.
- **Operative indications: articular step-off over 2-3 mm** (debated: stability and alignment matter more), **condylar widening over 5 mm**, **instability (over 10 degrees varus/valgus in extension)**, **medial plateau fractures** (inherently unstable), open fractures, compartment syndrome, vascular injury.
- **Principles:** restore the **mechanical axis and joint stability** first (more important than perfect articular congruity), elevate the joint surface, **support it with subchondral rafting screws and bone graft/substitute** (calcium phosphate cement resists subsidence best), **buttress** the metaphysis, repair menisci.

- **Approaches:** anterolateral (lateral plateau; submeniscal arthrotomy to see the joint); posteromedial (Lobenhoffer, between the medial gastrocnemius and pes anserinus/semimembranosus); posterior (Burks-Schaffer, Carlson); avoid a single midline incision with extensive stripping for bicondylar fractures (**dual incisions with a skin bridge over about 7 cm** historically; soft tissue is the priority).
- **Staged protocol for high-energy fractures: spanning external fixation** (ligamentotaxis), CT in the frame, definitive fixation once the soft tissues settle.
- **Arthroscopy-assisted reduction (ARIF)** for lateral split/depression fractures.
- **Elderly with severe comminution/arthritis:** acute TKA in selected cases.

Complications

- **Infection and wound breakdown** (bicondylar, early surgery through poor soft tissue), **post-traumatic arthritis** (malalignment and instability more than step-off), **stiffness/arthrofibrosis**, loss of reduction (subsidence), **compartment syndrome**, **peroneal nerve injury**, DVT, malunion.

2. Tibial shaft fractures (brief link)

- Commonest long-bone fracture; **highest rate of open fractures** and compartment syndrome.
- **Reamed intramedullary nail** for most displaced fractures (**SPRINT**: no difference between reamed and unreamed overall; possible benefit of reaming in closed fractures). **Suprapatellar (semi-extended) nailing** eases proximal and distal third fractures (less anterior knee pain debated).
- **Proximal third fractures** tend to **valgus and procurvatum (apex anterior)** with nailing: use **blocking (Poller) screws**, a more lateral/proximal start point, extended positioning, plates.
- **Distal third**: nail with blocking screws, or plate; fix the fibula if needed for alignment.
- Non-operative (Sarmiento functional bracing) for low-energy, stable fractures with acceptable alignment (under 5 degrees varus/valgus, under 10 degrees AP angulation, under 1 cm shortening, under 10 degrees rotation).
- **Anterior knee pain** is the commonest complication after tibial nailing.

3. Tibial pilon (plafond) fractures

Mechanism and features

- **Axial loading**: the talus is driven into the distal tibial articular surface (falls from height, road traffic accidents). Distinct from rotational ankle fractures.
- **Soft-tissue injury is the dominant problem** (thin anterior and medial soft tissue envelope).
- Fragments (Chaput's): typically **medial**, **anterolateral (Chaput, attached to AITFL)**, **posterolateral (Volkmann, attached to PITFL)**, with central die-punch impaction; associated fibula fracture in about 75%.
- **Associated injuries:** spine fractures, calcaneus fractures, contralateral injuries (falls from height).

Classification

Ruedi-Allgower	Description
I	Undisplaced cleavage fracture of the joint
II	Displaced articular fracture, minimal comminution
III	Comminuted and impacted articular surface

- **AO/OTA 43:** A extra-articular, B partial articular, C complete articular (C3 = comminuted articular + metaphyseal).
- **Tscherne** for closed soft-tissue injury (C0-C3).

Management: the staged protocol

1. **Stage 1 (urgent): spanning external fixator** (tibia to calcaneus/first metatarsal) to restore length, alignment and rotation by ligamentotaxis; ± **fibular fixation** (if the plafond team and soft tissue plan allow; malreduced fibula makes later tibial reduction impossible).
2. **CT in the frame**: plan approaches.
3. **Stage 2 (when soft tissues allow, usually 7-21 days; "wrinkle sign", blisters re-epithelialised): ORIF** of the articular surface with plates (anterolateral, anteromedial, posterolateral, medial approaches depending on fragments), bone graft for metaphyseal defects.
4. **Rüedi-Allgöwer principles:** (1) restore fibular length, (2) reconstruct the articular surface, (3) bone graft the metaphyseal defect, (4) buttress the

medial tibia.

5. **Alternatives:** definitive **circular (Ilizarov) or hybrid external fixation** with limited internal fixation when soft tissues are poor; **primary ankle arthrodesis** for unreconstructable fractures.
6. **Approaches: anterolateral** (between the EDL/peroneus tertius and the fibula; superficial peroneal nerve at risk), **anteromedial** (medial to tibialis anterior; thinnest soft tissue), **posterolateral** (between peroneals and FHL; sural nerve), **medial**. Skin bridges of at least 5-7 cm between incisions (debated).

Complications

- **Wound dehiscence and deep infection** (historically up to 30-40% with early ORIF; staged protocols reduced this to under 5-10%).
- **Post-traumatic ankle arthritis** (common; related to the initial cartilage damage more than reduction quality), malunion (varus), non-union (metaphyseal), stiffness, chronic pain, complex regional pain syndrome.

> Exam pearl: **Schatzker II (split-depression) is the commonest; IV (medial) = fracture-dislocation with popliteal artery and peroneal nerve risk; VI = metaphyseal-diaphyseal dissociation with compartment syndrome.** The **lateral meniscus** is the commonest associated soft-tissue injury.

> Exam pearl: **Pilon = axial load; soft tissue dictates timing; stage it: span, scan, plan, then fix (7-21 days).** Rüedi-Allgöwer's four steps start with **fibular length**.

Ankle fractures

1. Applied anatomy

- The ankle mortise: tibial plafond, medial malleolus, lateral malleolus (fibula, the main lateral restraint). The **talus is wider anteriorly** (stable in dorsiflexion).
- **Ramsey and Hamilton: 1 mm of lateral talar shift reduces tibiotalar contact area by about 42%** (the basis for anatomical reduction).
- **Medial ligament (deltoid): superficial** (crosses ankle and subtalar; tibionavicular, tibiospring, tibiocalcaneal) and **deep** (deep anterior and posterior tibiotalar; **the primary restraint to lateral talar shift**).
- **Lateral ligaments: ATFL** (weakest, commonest sprain; tight in plantarflexion), **CFL** (crosses ankle and subtalar joints), PTFL (strongest).
- **Syndesmosis: AITFL**, **PITFL** (strongest component, with its deep part the inferior transverse ligament), interosseous ligament, interosseous membrane.
- The posterior malleolus is attached to the PITFL; a posterior malleolar fracture can destabilise the syndesmosis.

2. Classifications

Lauge-Hansen (mechanism: position of the foot + direction of force)

Type	Frequency	Sequence of injury	Fibular fracture
Supination-external rotation (SER)	**Commonest (about 40-75%)**	I AITFL, II **short oblique/spiral fibula at the level of the syndesmosis** (anteroinferior to posterosuperior), III PITFL or posterior malleolus, IV medial malleolus fracture or deltoid rupture	Weber B
Supination-adduction (SAD)	About 10-20%	I transverse fibular avulsion below the joint (or lateral ligament rupture), II **vertical medial malleolus fracture** with **medial plafond impaction** (fix and elevate)	Weber A

Type	Frequency	Sequence of injury	Fibular fracture
Pronation-external rotation (PER)	About 7-20%	I medial malleolus/deltoid, II AITFL, III **high spiral fibula fracture above the syndesmosis**, IV PITFL/posterior malleolus	Weber C (includes **Maisonneuve**)
Pronation-abduction (PAB)	About 5-20%	I medial malleolus/deltoid, II syndesmosis/posterior malleolus, III **comminuted/transverse fibula fracture at or above the syndesmosis** (lateral comminution)	Weber B/C

- Remember: the **first word is the foot position**, the second the force; the **first injured structure is on the tension side** (supination = lateral first; pronation = medial first).

Danis-Weber (level of the fibular fracture relative to the syndesmosis)

- **A:** below the syndesmosis (infra-syndesmotic); syndesmosis intact.
- **B:** at the level of the syndesmosis (trans-syndesmotic); syndesmosis may be injured.
- **C:** above the syndesmosis; syndesmosis torn (needs assessment and usually fixation).

Special patterns

- **Maisonneuve fracture: proximal fibula fracture + syndesmotic and interosseous membrane disruption + medial injury** (PER). Always palpate the whole fibula in an "isolated" medial malleolar fracture or deltoid injury. Treat with **syndesmotic fixation** (fibula itself usually not fixed).
- **Bosworth fracture-dislocation:** fibula **trapped behind the posterolateral tibia (incisura)**; irreducible by closed means: **emergency open reduction**.
- **Tillaux-Chaput** (AITFL avulsion from the tibia) and **Wagstaffe-Le Fort** (AITFL avulsion from the fibula) fragments.
- **Posterior malleolus** fractures: **Haraguchi classification** (CT): I posterolateral oblique (commonest), II medial extension, III small shell (rim). **Bartoníček-Rammelt** classification relates to the incisura.

3. Assessment

- **Ottawa ankle rules** (radiograph if pain in the malleolar zone plus tenderness at the posterior edge/tip of either malleolus, or inability to bear weight four steps immediately and in the department; foot series if tenderness at the base of the fifth metatarsal or navicular).
- **Radiographs:** AP, lateral, **mortise view** (15-20 degrees internal rotation).
  - **Medial clear space: over 4-5 mm** (or greater than the superior clear space) = **deltoid injury / lateral talar shift**.
  - **Tibiofibular clear space** (AP, 1 cm above the plafond): **over 5-6 mm = syndesmotic injury** (most reliable radiographic measure).
  - **Tibiofibular overlap:** under 6 mm on AP or under 1 mm on mortise = syndesmotic injury.
  - **Talocrural angle** (about 83 degrees ±4): assesses fibular length; **dime sign** (continuous curve between the fibular tip and lateral talus) shows fibular length.
- **Stress views: gravity or manual external rotation stress** (medial clear space over 5 mm) or **weight-bearing radiographs** decide whether an isolated Weber B (SER) fracture is stable: **a weight-bearing mortise with a normal medial clear space indicates stability** even if the gravity stress view widens.
- **CT** for posterior malleolar fractures, pilon-type impaction, and syndesmosis assessment; **MRI** for occult injuries.

4. Management

Stable injuries (non-operative)

- **Weber A**, **isolated undisplaced Weber B with a stable mortise** (normal medial clear space on weight-bearing film), isolated undisplaced medial malleolar fractures.
- Treatment: **walking boot or cast with early weight bearing as tolerated** for 6 weeks. The **AIM trial** (elderly, unstable ankle fractures, close-contact casting vs ORIF) showed **equivalent function at 6 months** with skilled casting; **CROSSBAT** (Kortekangas): undisplaced/minimally displaced Weber B can be treated non-operatively (casting for 3 weeks equals 6 weeks).

Unstable injuries (operative)

- **Bimalleolar**, **trimalleolar**, unstable SER IV equivalents (fibula + deltoid with talar shift), syndesmotic injuries, **open fractures**, irreducible dislocations, **talar shift**.
- **Emergency: reduce a dislocated ankle immediately** (skin and neurovascular compromise), splint, elevate; ORIF when swelling allows (within about 24 hours or after 7-10 days).
- **Fibula:** restore **length and rotation**: lag screw with **one-third tubular neutralisation plate (lateral)** or **posterolateral antiglide plate** (Weber B; biomechanically stronger, less hardware prominence, risk to peroneal tendons distally). Intramedullary fibular nail in poor soft tissue/elderly.
- **Medial malleolus: two 4.0 mm partially threaded cancellous screws** perpendicular to the fracture, or tension-band wiring for small fragments; **vertical (SAD) fractures need horizontal screws or an antiglide plate**, plus elevation of medial plafond impaction. The **deltoid ligament usually heals without repair** once the fibula is reduced (repair if the medial clear space does not reduce: interposed deltoid or tibialis posterior tendon; some advocate repair in high-demand athletes).
- **Posterior malleolus:** fix when **over about 25-33% of the articular surface** (traditional rule), when there is **posterior talar subluxation**, an **intercalary/impacted fragment**, or when it destabilises the syndesmosis. Current evidence favours **direct fixation (posterolateral approach) of larger fragments** because it restores syndesmotic stability and incisura congruity better than a syndesmotic screw.
- **Syndesmosis:**
  - **Intraoperative tests: Cotton (hook) test** (lateral pull on the fibula with a bone hook: over about 2 mm widening), external rotation stress.
  - **Fixation: screw(s) (3.5 or 4.5 mm, through 3 or 4 cortices, 2-5 cm above the joint, parallel to the plafond, angled about 25-30 degrees anteriorly from posterior)** or **suture-button (dynamic) fixation**. **Suture-buttons** give fewer malreductions and less need for removal (better functional scores in some trials); screws may be removed after about 3 months (debated).
  - **Accurate reduction is critical**: **direct visualisation of the incisura** or intraoperative CT, because **malreduction (up to about 30-50% on CT)** is the main cause of poor outcome. Clamp along the transmalleolar axis to avoid over-compression or malrotation.

- **Post-operative:** early range of motion; weight bearing in a boot from about 2-6 weeks (early weight bearing is safe in most fixed fractures).

Special groups

- **Diabetic patients (especially with neuropathy):** high complication rates (infection, Charcot, loss of fixation). **"Super-construct" fixation** (longer plates, multiple syndesmotic screws, transarticular pins), prolonged non-weight bearing (about double the usual time).
- **Elderly osteoporotic:** locking plates, fibular nails, **tibiotalocalcaneal (hindfoot) nail** for very poor bone/soft tissue; skilled close-contact casting (AIM).
- **Open fractures:** debridement, antibiotics, stabilisation (external fixation or internal fixation depending on contamination), soft-tissue cover.

5. Ankle sprains and related injuries

- **Lateral ankle sprain:** ATFL then CFL. **Anterior drawer** (ATFL), **talar tilt** (CFL). **Functional treatment** (brace, early motion, proprioceptive rehabilitation) for all grades acutely.
- **Chronic lateral ankle instability** after failed rehabilitation: **anatomical repair (Broström-Gould: ATFL/CFL repair reinforced with the inferior extensor retinaculum)**; anatomical reconstruction with tendon graft for poor tissue, failed repair, generalised laxity or high BMI; non-anatomical (Chrisman-Snook, Watson-Jones, Evans) rarely now (stiffness of the subtalar joint).
- **High ankle sprain (syndesmosis): squeeze test**, external rotation (Kleiger) test; longer recovery; stabilise if unstable on stress/arthroscopy.
- **Missed injuries around a "sprain": lateral process of the talus fracture** (snowboarder's), **anterior process of the calcaneus fracture**, **base of fifth metatarsal fracture**, **osteochondral lesion of the talus**, **peroneal tendon subluxation/tear**, **Maisonneuve fracture**.
- **Osteochondral lesions of the talus: anterolateral** (shallow, wafer-shaped, traumatic) vs **posteromedial** (deeper, cup-shaped, often atraumatic). **Berndt-Harty** staging; MRI/CT; **arthroscopic debridement and microfracture/bone-marrow stimulation for lesions under about 1.5 cm^2 (or under 15 mm diameter)**; osteochondral autograft (OATS) or other cartilage procedures for larger lesions.

6. Complications of ankle fractures

- **Post-traumatic arthritis** (malreduction of fibular length/rotation, syndesmotic malreduction, articular damage).
- **Wound complications and infection** (smokers, diabetes, early surgery through swollen skin).
- **Syndesmotic malreduction**, **broken syndesmotic screws** (usually harmless), loss of reduction.
- **Superficial peroneal nerve** injury (lateral incision, about 7 cm above the tip of the fibula it pierces the fascia), sural nerve (posterolateral approach), saphenous nerve (medial).
- Complex regional pain syndrome, stiffness, DVT.

> Exam pearl: **SER is the commonest Lauge-Hansen type (fibula at the syndesmosis, anteroinferior to posterosuperior). PER IV gives a high fibular fracture; Maisonneuve = proximal fibula + syndesmotic disruption. 1 mm talar shift = 42% loss of contact area.**

> Exam pearl: **Medial clear space over 4-5 mm = deltoid injury; tibiofibular clear space over 5-6 mm = syndesmotic injury.** Weight-bearing radiographs decide stability of an isolated Weber B. **Syndesmotic malreduction is the main cause of poor outcome; visualise the incisura.**

Proximal humerus and shoulder girdle

1. Proximal humerus fractures

Anatomy

- Four parts (**Codman**): **head** (articular segment), **greater tuberosity** (supraspinatus, infraspinatus, teres minor), **lesser tuberosity** (subscapularis), **shaft**.
- **Head blood supply:** the **posterior humeral circumflex artery** supplies most of the head (Hettrich, 2010; about 64%); the **anterior humeral circumflex artery** and its **arcuate artery (of Laing)**, entering at the bicipital groove, was classically considered the main supply.
- **Hertel's predictors of head ischaemia: metaphyseal extension under 8 mm** attached to the head (posteromedial calcar), **disrupted medial hinge (over 2 mm displacement)**, and the **anatomical neck fracture** pattern. Head ischaemia does not always mean AVN.
- **Deforming forces:** greater tuberosity pulled **superiorly and posteriorly** (supraspinatus/infraspinatus); lesser tuberosity **medially** (subscapularis); shaft **medially and anteriorly** (pectoralis major).
- **Axillary nerve** runs about **5-7 cm below the acromion** (deltoid-splitting approach danger zone); the **arcuate artery** and **cephalic vein** in the deltopectoral approach.

Neer classification

- A "part" is displaced if **over 1 cm displacement or over 45 degrees angulation** (for the greater tuberosity, **over 5 mm displacement** is significant in young/overhead patients, as superior displacement causes impingement).
- One-part (undisplaced, about 80%), two-part (surgical neck commonest two-part; greater tuberosity; anatomical neck rare), three-part (surgical neck + one tuberosity), four-part (both tuberosities + surgical neck), **fracture-dislocations** and **head-splitting** fractures.
- **Valgus-impacted four-part fracture:** better prognosis (medial periosteal hinge preserved, lower AVN).
- **AO/OTA 11** classification also used.

Management

- **Non-operative** (sling, early pendulum exercises at about 2 weeks): most fractures, especially **one-part** and in the **elderly**. The **ProFHER trial** (2015, elderly displaced surgical-neck-involving fractures): **no benefit of surgery over non-operative care** at 2 and 5 years.
- **Isolated greater tuberosity fracture: fix if displaced over 5 mm** (screws, suture-bridge/tension band) in active patients.
- **Two/three-part fractures in active patients: proximal humeral locking plate** (deltopectoral or deltoid-split approach). **Key technical points:** restore the **medial calcar support** (an inferomedial "calcar screw"; medial comminution predicts varus collapse), avoid **plate placed too high** (subacromial impingement; plate about 8 mm below the top of the greater tuberosity), **avoid screw cut-out** (commonest complication: primary or secondary intra-articular screw penetration with varus collapse), use **tuberosity sutures** through the cuff. Fibular strut allograft in the canal for medial support.
- **Intramedullary nails** for two-part surgical neck fractures (straight nails reduce cuff damage).
- **Four-part fractures, head-split, fracture-dislocations in the elderly:**
  - **Reverse total shoulder arthroplasty (RTSA)** is now preferred in elderly patients (over about 70): **more predictable function**, does not depend on tuberosity healing (though tuberosity repair improves rotation).
  - **Hemiarthroplasty** in younger patients where tuberosity healing is likely: outcome **depends on tuberosity healing** (correct height and version: top of the head about 5-8 mm above the greater tuberosity, retroversion about 20-30 degrees).
  - **ORIF** in young patients (attempt to preserve the head; AVN may still occur).
- **Complications: AVN** (four-part and head-split, about 10-30%+), malunion (varus, tuberosity), non-union (surgical neck), **screw cut-out**, stiffness, **axillary nerve injury**, infection.

2. Shoulder dislocation (acute, trauma aspect)

- **Anterior (about 95-97%):** abduction-external rotation; associated **Bankart lesion** (anteroinferior labrum/capsule), **Hill-Sachs** (posterosuperolateral

head impaction), **greater tuberosity fracture** (in older patients; often reduces with the head), **axillary nerve injury** (commonest nerve), **rotator cuff tear** in patients over 40 (**test cuff strength after reduction; MRI/ultrasound if weak**).

- **Recurrence after first dislocation** is inversely related to age: **up to about 90% under 20 years**, much lower over 40. Arthroscopic Bankart repair after first dislocation lowers recurrence in young athletes.
- **Posterior: seizures, electrocution**, **lightbulb sign**, **trough (reverse Hill-Sachs/McLaughlin) sign**, inability to externally rotate; **often missed** (always get an **axillary lateral** view). Reverse Hill-Sachs over about 25% of the head: McLaughlin procedure (subscapularis/lesser tuberosity transfer); over about 50%: arthroplasty/allograft.
- **Luxatio erecta** (inferior): arm locked in abduction; neurovascular injury common.

3. Clavicle fractures

- Commonest fracture of the shoulder girdle; **middle third (about 80%)** (Allman group I), lateral third (about 15%, group II), medial third (about 5%, group III).
- **Midshaft:** most heal non-operatively. **Displaced/shortened midshaft fractures** (100% displacement, **shortening over about 2 cm**, comminution) have higher **non-union (about 15%)** and **symptomatic malunion** rates without surgery (**COTS trial**, 2007: **plate fixation lowered non-union and improved function** in displaced fractures; later trials show smaller differences). **Absolute indications: open fracture, skin tenting with impending perforation, neurovascular injury**, floating shoulder (relative), polytrauma.
- **Fixation:** superior or anteroinferior plate (anteroinferior: less prominence, longer screws, less subclavian vessel risk) or intramedullary devices.
- **Lateral third (Neer): Type I** (lateral to the coracoclavicular ligaments, stable), **Type II** (medial to CC ligaments; the medial fragment detaches from the CC ligaments, **high non-union (about 30%)**: **fix** with hook plate, lateral locking plate or CC suspension), **Type III** (intra-articular into AC joint; late AC arthritis), IV (children, periosteal sleeve), V (comminuted).

- **Medial clavicle / sternoclavicular:** posterior SC dislocation endangers great vessels and trachea (**CT; reduce in theatre with a thoracic surgeon available**); medial physis closes at about 23-25 years (physeal injuries mimic dislocations).

4. Acromioclavicular (AC) joint injuries

- **Rockwood classification:**

Type	AC ligament	CC ligament	Displacement	Treatment
I	Sprained	Intact	None	Sling
II	Torn	Sprained	Up to 25% (subluxed)	Sling
III	Torn	Torn	**25-100%** CC distance increase	**Non-operative first** in most; surgery for heavy labourers/overhead athletes or failed rehab
IV	Torn	Torn	**Clavicle posterior** through trapezius (axillary view)	Surgery
V	Torn	Torn	**100-300%** (deltotrapezial fascia stripped)	Surgery
VI	Torn	Torn	Clavicle **inferior** (subacromial or subcoracoid)	Surgery (rare)

- **Zanca view** (10-15 degrees cephalic tilt) for the AC joint; **normal CC distance about 11-13 mm**.
- Surgery: anatomical CC ligament reconstruction (graft), suture-button suspension, hook plate (acute), **Weaver-Dunn** (coracoacromial ligament transfer, historical).

5. Scapula fractures

- High-energy injuries; **associated injuries in about 80-90%** (rib fractures, pneumothorax, pulmonary contusion, head and spine injury).

- **Body fractures** (commonest): mostly non-operative. Operate for significant **medialisation of the glenoid (over 20 mm)**, **angulation over 45 degrees**, **glenopolar angle under 22 degrees**.
- **Glenoid fractures (Ideberg):** fix if **articular step-off over 4-5 mm**, or rim fractures causing **instability (over about 20-25% of the glenoid)**.
- **Floating shoulder:** ipsilateral clavicle and glenoid neck fractures (superior shoulder suspensory complex, **Goss**: double disruption of the SSSC). Fix one or both when displaced.
- **Scapulothoracic dissociation:** lateral displacement of the scapula (**scapular index**), often with **brachial plexus avulsion and subclavian/axillary artery injury**; high mortality; flail arm, often amputation.

6. Humeral shaft fractures

- **Holstein-Lewis fracture:** spiral fracture of the distal third with **radial nerve** entrapment risk.
- **Non-operative (Sarmiento functional brace)** is standard: acceptable alignment **under 20 degrees anterior angulation, under 30 degrees varus, under 3 cm shortening**. **FISH trial** (2021) and others: surgery gives faster early recovery but similar long-term function; non-union about 10-15% with bracing (proximal third and transverse patterns more).
- **Operative indications: open fracture**, **vascular injury**, **floating elbow**, polytrauma (to allow crutch use), **pathological fracture**, bilateral fractures, failed bracing, segmental fractures, intra-articular extension, brachial plexus injury.
- **Plating** (anterolateral, posterior or MIPO) is preferred over **antegrade nailing** (nailing causes more **shoulder pain** and reoperation; useful in pathological fractures).
- **Radial nerve palsy with a closed fracture** (about 10-18%): mostly **neurapraxia**: **observe** (about 70-90% recover spontaneously; wait about 3-4 months, then EMG; explore if no recovery). **Explore early** for **open fractures** (nerve may be lacerated) and palsy with an **open or high-energy fracture needing fixation**. **Palsy after closed manipulation** is debated (traditionally explore). Tendon transfers (pronator teres to ECRB, FCR/FCU to EDC, palmaris longus to EPL) if no recovery.

Exam pearl: **Neer part = over 1 cm or 45 degrees (greater tuberosity over 5 mm). ProFHER: elderly displaced fractures do as well without surgery. Reverse arthroplasty is preferred for four-part fractures in the elderly; hemiarthroplasty depends on tuberosity healing. Medial calcar support prevents varus collapse after locking plates.**

Exam pearl: **Posterior shoulder dislocation (seizure, electrocution) = lightbulb sign, missed without an axillary view. Neer II lateral clavicle fracture has a high non-union rate: fix. Rockwood III AC injury: non-operative first. Radial nerve palsy with closed humeral shaft fracture: observe.**

Distal radius and carpus

1. Distal radius fractures

Normal radiographic parameters

Parameter	Normal	Acceptable after reduction (active adult)
Radial height (length)	About 11-12 mm	Shortening **under 3-5 mm**
Radial inclination	About 22-23 degrees	**Over about 15 degrees**
Volar (palmar) tilt	About 11 degrees (range 0-20)	**Dorsal tilt under 10 degrees** (neutral or better)
Ulnar variance	Neutral (about 0 ±2 mm)	**Under +2-3 mm** ulnar positive
Articular step-off / gap	0	**Under 2 mm**

- **Teardrop angle** (lateral) about 70 degrees; **AP distance** of the lunate facet.
- **Instability predictors (Lafontaine): dorsal angulation over 20 degrees, dorsal comminution, intra-articular involvement, associated ulnar fracture, age over 60**. Three or more = likely loss of reduction in a cast. Also initial **shortening over 5 mm**.

Eponyms and classifications

- **Colles':** extra-articular, **dorsal angulation and displacement**, radial shortening ("dinner-fork"), fall on outstretched hand.
- **Smith's: volar** angulation ("reverse Colles"; garden-spade deformity); **unstable**: needs **volar buttress plate**.
- **Barton's: intra-articular rim fracture with carpal subluxation** (volar Barton commoner): **buttress plate**.
- **Chauffeur's: radial styloid fracture** (associated with scapholunate injury/perilunate).
- **Die-punch:** depressed **lunate facet** fracture.
- **Frykman** (I-VIII: radiocarpal/DRUJ involvement, ulnar styloid), **Fernandez** (mechanism-based I bending, II shearing, III compression, IV avulsion, V

combined), **AO/OTA 23 A/B/C**, **Melone** (four-part intra-articular).

- **Volar lunate facet ("critical corner") fragment:** small and distal; if not captured, **volar carpal subluxation** follows; needs plate placement distal to the watershed line or a fragment-specific plate/hook.

Management

- **Undisplaced or acceptable alignment:** cast or splint 4-6 weeks; check radiographs weekly for 2-3 weeks.
- **Displaced: closed reduction** (haematoma block, Bier block) and cast; **re-displacement** is common in unstable patterns.
- **Operative indications (young/active):** failed closed reduction, **unstable patterns**, articular step-off over 2 mm, **volar shearing (Smith's, Barton's)**, open fractures, associated carpal injuries, bilateral.
- **Implants: volar locking plate** (most common; **flexor carpi radialis (modified Henry) approach**; plate **proximal to the watershed line** to avoid **FPL rupture**), K-wires (DRAFFT trial: no difference between K-wires and volar plating at 12 months in dorsally displaced fractures), dorsal plates (dorsal comminution, dorsal shear; extensor irritation), fragment-specific fixation, bridging plates (dorsal spanning) and external fixation (comminuted, polytrauma).
- **Elderly (over 65):** non-operative treatment gives **similar functional outcomes** despite worse radiographs (WRIST trial and others: volar plating vs casting in older adults gave small early differences; **AAOS** guidance: weak evidence for surgery over 65). Volar plate for active older patients with unacceptable displacement.

Complications

- **Median nerve dysfunction / acute carpal tunnel syndrome** (progressive symptoms: **urgent carpal tunnel release** with fixation).
- **EPL rupture** after **undisplaced** fractures (attrition at **Lister's tubercle**, ischaemia; about 3-8 weeks later): **EIP to EPL transfer**. After volar plating, **long dorsal screws** can rupture EPL/EDC (use shorter screws, skyline/dorsal tangential view).
- **FPL rupture** with volar plates placed **distal to the watershed line** (plate prominence): remove plate after union.

- **Malunion** (dorsal tilt, shortening: **corrective osteotomy with bone graft**; ulnar positive variance: **ulnar shortening osteotomy**).
- **Complex regional pain syndrome (CRPS): vitamin C 500 mg daily for 50 days** may reduce incidence (AAOS moderate evidence, though conflicting trials). Budapest criteria.
- **Stiffness**, DRUJ instability/arthritis, **ulnar impaction**, post-traumatic arthritis (articular step-off over 2 mm), median/ulnar nerve injury, tendon irritation.
- **Ulnar styloid fractures: base fractures** may be associated with DRUJ instability; fixation rarely needed (test DRUJ after radius fixation).

2. Scaphoid fractures

Anatomy

- About 80% covered with articular cartilage. **Retrograde blood supply** from the **dorsal carpal branch of the radial artery** entering at the **dorsal ridge (waist)**, supplying about 70-80% (including the **proximal pole**); the **superficial palmar branch** supplies the distal tubercle. Hence **proximal pole fractures** have the highest **AVN and non-union** rates.

Clinical and imaging

- Fall on the outstretched hand; **anatomical snuffbox tenderness**, **scaphoid tubercle tenderness**, **pain on axial thumb compression** (combination increases sensitivity).
- **Scaphoid series** (PA in ulnar deviation, lateral, obliques). Initial radiographs miss about 20%.
- **Suspected fracture with normal radiographs: MRI** (best; sensitivity near 100%), or CT; otherwise immobilise and re-image at 10-14 days.
- **Displacement: over 1 mm**, **intrascaphoid angle over 35 degrees** (lateral, **humpback deformity**), **height-to-length ratio over 0.65**, **DISI** (radiolunate angle over 15 degrees).

Classification and treatment

Location	Frequency	Treatment
Distal pole / tubercle	About 10-20%	Cast 4-6 weeks; heals well

Location	Frequency	Treatment
Waist	**About 65-80% (commonest)**	**Undisplaced: cast** (below-elbow cast without the thumb is as effective as including it) 6-10 weeks, **or percutaneous screw** (faster return to work/sport; **SWIFFT trial** 2020: **no long-term benefit of early fixation** for undisplaced/minimally displaced (under 2 mm) waist fractures; treat with cast and fix if non-union). **Displaced (over 1-2 mm): fix (headless compression screw)**
Proximal pole	About 10-20%	**Fix** (high non-union and AVN risk); **dorsal approach**

- Classifications: **Herbert** (A stable acute: A1 tubercle, A2 incomplete waist; B unstable acute: B1 distal oblique, B2 complete waist, B3 proximal pole, B4 transscaphoid perilunate; C delayed union; D non-union), **Russe** (fracture line: horizontal oblique, transverse, vertical oblique), Mayo.
- **Screw position:** central in the scaphoid (longest screw in the **central axis** gives most stability); **volar approach** for waist/distal fractures (percutaneous through the trapezium edge), **dorsal approach** for proximal pole.

Scaphoid non-union

- **Risk factors: proximal pole**, **displacement over 1 mm**, delayed diagnosis/treatment, smoking, instability.
- **Natural history: SNAC wrist (scaphoid non-union advanced collapse)**: radial styloid-scaphoid arthritis first, then capitate-scaphoid, then lunocapitate (**radiolunate joint spared**).
- **Treatment: non-union without arthritis**: ORIF with **bone graft**: **volar wedge graft (Fisk-Fernandez)** for humpback deformity; **vascularised bone graft (1,2-ICSRA, Zaidemberg; medial femoral condyle free flap)** for **AVN of the proximal pole** (absent punctate bleeding). **SNAC:** radial styloidectomy (stage I), **proximal row carpectomy** (needs a healthy capitate head and lunate fossa) or **scaphoid excision + four-corner fusion** (stage II-III), total wrist fusion.
- **Preiser disease:** idiopathic AVN of the scaphoid.

3. Other carpal fractures

- **Triquetrum** (second commonest carpal fracture): **dorsal chip** (avulsion/impaction): symptomatic treatment.
- **Hook of hamate:** golfers, baseball, racquet sports; **carpal tunnel view** or **CT**; **ulnar nerve (Guyon) symptoms**, **FDP tendon ruptures (ring/little)**; **excision** of the hook for non-union/symptoms.
- **Pisiform**, **trapezium** (ridge fractures; FCR), **capitate** (scaphocapitate syndrome: head rotated 180 degrees).
- **Lunate: Kienböck disease** (AVN of the lunate): **ulnar negative variance** association; **Lichtman staging** (I normal radiograph/MRI changes, II sclerosis, **IIIA collapse with normal carpal alignment**, **IIIB fixed scaphoid rotation (ring sign, radioscaphoid angle over 60 degrees)**, IV pancarpal arthritis). **Joint levelling (radial shortening osteotomy)** for ulnar-negative stage I-IIIA, revascularisation (4,5-ECA graft), **proximal row carpectomy or STT/SC fusion for IIIB**, wrist fusion/arthroplasty for IV.

4. Carpal instability and perilunate injuries (trauma aspect)

- **Mayfield progression (perilunate instability)** with wrist extension, ulnar deviation and intercarpal supination: **I scapholunate disruption, II lunocapitate, III lunotriquetral, IV lunate dislocation** (volar, "**spilled teacup**" on lateral).
- **Greater arc injuries** (through bones: **transscaphoid perilunate fracture-dislocation**, commonest) vs **lesser arc** (purely ligamentous).
- **Signs:** broken **Gilula's arcs** (three smooth arcs on PA), **triangular lunate** on PA, **spilled teacup** on lateral, capitate dorsal to the lunate.
- **Treatment: urgent closed reduction** (Tavernier manoeuvre), then **open reduction, ligament repair and internal fixation** (dorsal ± volar approach); acute **median nerve** symptoms need urgent decompression. Up to 25% are **missed** initially.
- **Scapholunate dissociation: Terry Thomas sign** (SL gap **over 3 mm**), **scaphoid ring (cortical ring) sign**, **scapholunate angle over 60-70 degrees** (normal 30-60), **DISI** (lunate extended). **Watson's scaphoid shift test.** Acute: repair; chronic reducible: reconstruction (e.g. three-ligament tenodesis); **SLAC wrist** progression (radial styloid, then radioscaphoid, then capitolunate; radiolunate spared).

- **Lunotriquetral injury: VISI** (lunate flexed), ballottement (Reagan) test, shuck test.

5. Distal radioulnar joint and TFCC

- **TFCC** components: articular disc, dorsal and volar radioulnar ligaments (primary DRUJ stabilisers), meniscal homologue, ECU subsheath, ulnocarpal ligaments (ulnolunate, ulnotriquetral).
- **Palmer classification: Class 1 traumatic** (1A central perforation: commonest, debride; **1B peripheral ulnar detachment** (with or without ulnar styloid fracture): repair; 1C distal ulnocarpal ligaments; 1D radial detachment), **Class 2 degenerative** (ulnar impaction, stages A-E).
- **Ulnar impaction syndrome:** ulnar positive variance, lunate/ulnar head chondromalacia; **ulnar shortening osteotomy** or wafer procedure.
- **DRUJ instability tests:** piano-key sign, ballottement in neutral, pronation and supination.

> Exam pearl: **Acceptable distal radius: radial shortening under 3-5 mm, dorsal tilt under 10 degrees, inclination over 15 degrees, step-off under 2 mm.** Elderly over 65: non-operative results are comparable. **EPL rupture after undisplaced fractures; FPL rupture after plates beyond the watershed line.**

> Exam pearl: **Scaphoid: retrograde blood supply, proximal pole AVN; MRI for occult fracture; SWIFFT: cast undisplaced waist fractures; fix displaced and proximal pole fractures.** SNAC/SLAC both **spare the radiolunate joint**.

> Exam pearl: **Perilunate: Mayfield I-IV (SL to lunate dislocation); transscaphoid perilunate is the commonest greater-arc injury; spilled teacup = lunate dislocation.**

Elbow and forearm

1. Distal humerus fractures (adult)

- **Two-column construct:** medial and lateral columns joined by the trochlea (the "tie-arch"). Most are **intra-articular (AO/OTA 13C)**.
- Classifications: **AO/OTA 13 A (extra-articular), B (partial articular: lateral or medial condyle, coronal shear of capitellum/trochlea), C (complete articular: C1 simple, C2 metaphyseal comminution, C3 articular comminution)**; Jupiter and Mehne (high/low T, Y, H, lambda patterns).
- **Treatment: ORIF with two plates** (orthogonal: posterolateral + medial, or **parallel: medial + lateral**, the latter biomechanically stronger), restoring articular surface, **columns and trochlear width**. O'Driscoll principles: as many long screws as possible in the distal fragments, screws passing through a plate and interdigitating, compression at the metaphysis.
- **Approach: posterior**, with **olecranon osteotomy** (**chevron, apex distal**, at the bare area (sigmoid notch), best articular exposure; risk of non-union/hardware irritation), **triceps-sparing (Alonso-Llames)**, **triceps-splitting**, or **triceps-reflecting (Bryan-Morrey, TRAP)**. **Identify and protect the ulnar nerve** (transposition is debated; many transpose anteriorly).
- **Elderly osteoporotic, comminuted, unreconstructable (C3): total elbow arthroplasty** (linked, semi-constrained; **McKee trial**: better early function than ORIF) with **lifelong lifting restrictions** (roughly 1 kg repeatedly, about 5 kg as a single lift).
- **Coronal shear fractures (capitellum/trochlea):** Bryan-Morrey (I Hahn-Steinthal: large capitellar fragment; II Kocher-Lorenz: articular shell; III comminuted; IV McKee: capitellum + trochlea), Dubberley. **Double-arc sign** on lateral radiograph. Fix with **posterior-to-anterior or headless compression screws**.
- **Complications: stiffness** (commonest), **ulnar neuropathy**, heterotopic ossification, non-union (supracondylar), olecranon osteotomy non-union, infection.

2. Elbow dislocation and instability

- **Commonest adult major joint dislocation after the shoulder; posterolateral is commonest.** Mechanism: axial load, valgus, supination

("**Horii circle**": ligament failure progresses **from lateral to medial**: LUCL first, then anterior and posterior capsule, then the MCL; O'Driscoll stages).

- **Simple dislocation** (no fracture): reduce, check stability through the arc of motion, **early active motion within 1-2 weeks** (prolonged immobilisation causes stiffness; avoid over 3 weeks). Hinged brace if unstable at 30-45 degrees of extension.
- **Complex dislocation** (with fractures).
- **"Terrible triad": posterior dislocation + radial head fracture + coronoid fracture** (with LCL disruption). Treatment sequence (lateral approach, inside out): **(1) coronoid fixation (or anterior capsule repair), (2) radial head fixation or replacement (never excise), (3) LCL (LUCL) repair to the lateral epicondyle**, then **(4) MCL repair or hinged external fixator** if still unstable.
- **Varus posteromedial rotatory instability: anteromedial facet fracture of the coronoid** + LCL tear (O'Driscoll coronoid type 2); fix the anteromedial facet with a buttress plate.
- **Posterolateral rotatory instability (PLRI): LUCL insufficiency** (after dislocation, or iatrogenic after lateral surgery/tennis elbow release). Tests: **lateral pivot-shift (O'Driscoll)** (supination, valgus, axial load, flexion: reduction clunk at about 40 degrees), posterolateral drawer, chair push-up, table-top relocation. Treat with **LUCL reconstruction** (palmaris/gracilis graft).
- **Coronoid fractures: Regan-Morrey I (tip), II (up to 50%), III (over 50%)**; **O'Driscoll: tip, anteromedial facet, basal**.

3. Olecranon fractures

- Intra-articular, pulled by the triceps. **Mayo classification: I undisplaced, II displaced stable (ulnohumeral joint stable), III unstable** (A non-comminuted, B comminuted). **Schatzker** and Colton also used.
- **Treatment:** undisplaced: splint and early motion. Displaced **transverse, non-comminuted**: **tension-band wiring** (two parallel K-wires (ideally engaging the anterior ulnar cortex) + figure-of-eight wire; converts triceps tension into articular compression). **Oblique, comminuted, distal to the coronoid, or Monteggia-type**: **plate fixation** (dorsal precontoured plate). **Elderly, low-demand**: non-operative even if displaced (good function), or excision with triceps advancement (up to about 50% of the olecranon if the coronoid and anterior soft tissues are intact).

- **Complications: symptomatic hardware (commonest; removal in up to about 50-80% with TBW)**, K-wire migration, ulnar nerve symptoms, loss of extension, arthritis.

4. Radial head and neck fractures

- **Mason classification (with Johnston and Morrey modifications):**

Type	Description	Treatment
I	Undisplaced or under 2 mm	**Early mobilisation** (aspiration of haemarthrosis + local anaesthetic relieves pain and allows testing for mechanical block)
II	**Displaced over 2 mm**, partial articular (one fragment)	Non-operative if no block and stable; **ORIF (headless screws, buried in the "safe zone")** if mechanical block or displacement
III	**Comminuted, whole head**	**Radial head replacement (arthroplasty)** if unreconstructable (over 3 fragments: ORIF results poor)
IV	With **elbow dislocation**	Fix or replace radial head + ligament repair

- **Safe zone for implants:** the **non-articular arc of about 90-110 degrees** that does not contact the ulna (proximal radioulnar joint): **between radial styloid and Lister's tubercle in neutral rotation** (the lateral arc in neutral).
- **Radial head is a secondary stabiliser to valgus stress** (the MCL is primary) and an important **longitudinal stabiliser** (with the interosseous membrane).
- **Never excise** the radial head when there is associated **MCL injury, coronoid fracture, interosseous membrane injury (Essex-Lopresti)** or elbow instability.
- **Radial head arthroplasty: avoid overstuffing** (gap in lateral ulnohumeral joint, capitellar wear, pain, loss of flexion): the top of the implant should be at the level of the proximal edge of the lesser sigmoid notch (about 1 mm below the coronoid tip). Metal monopolar stems (loose fit) common.
- **Essex-Lopresti injury:** radial head fracture + **interosseous membrane disruption** + **DRUJ disruption**: proximal migration of the radius (wrist pain). **Preserve or replace the radial head**; stabilise DRUJ; IOM reconstruction for chronic cases.

- **Posterior interosseous nerve** is at risk with lateral approaches: **pronate the forearm** (moves the PIN anteriorly and medially away from the field) in the **Kocher (anconeus and ECU interval)** and Kaplan (ECRB and EDC) approaches.

5. Forearm shaft fractures (adult)

- **Both-bone fractures** in adults need **ORIF with compression plates (3.5 mm LC-DCP)** restoring length, rotation and the **radial bow** (critical for forearm rotation; the **maximal radial bow** is about 60% along the radius from the proximal end).
- **Approaches: Henry (volar)** for the radius (**between brachioradialis and FCR distally; brachioradialis and pronator teres proximally**; superficial radial nerve and radial artery at risk; supinate to expose the supinator and protect the PIN proximally), **Thompson (dorsal)** between ECRB and EDC (PIN at risk proximally), ulna through the subcutaneous border (between ECU and FCU).
- **Monteggia (adult): proximal ulna fracture + radial head dislocation**. **Bado:**
  - **I** anterior dislocation, apex-anterior ulna (commonest in children).
  - **II posterior** dislocation, apex-posterior ulna (**commonest in adults**, often with radial head fracture: "**posterior Monteggia**").
  - III lateral dislocation (ulnar metaphysis).
  - IV anterior dislocation with both-bone fracture.
  - **Treatment:** anatomical **ulnar fixation (plate)**, which usually reduces the radial head; if not, open reduction (interposed annular ligament). **PIN palsy** (commonest nerve, Bado I/III) usually recovers.
- **Galeazzi: distal-third radial shaft fracture + DRUJ disruption** ("**fracture of necessity**": needs ORIF). Radial fractures **within about 7.5 cm of the articular surface** are more likely to have DRUJ instability. After plating, check the **DRUJ in supination**: stable = early motion; reducible but unstable = **pin radius to ulna in supination** or repair TFCC; irreducible = open (ECU tendon interposition).
- **Nightstick (isolated ulnar shaft) fracture:** non-operative if **under 50% displacement and under 10 degrees angulation**; otherwise plate.

- **Complications: synostosis** (single incision for both bones, high-energy, head injury, delayed surgery; excise after about 6-12 months), **refracture after plate removal** (avoid removal; protect 6 weeks), non-union, compartment syndrome, PIN injury.

Exam pearl: **Terrible triad = posterior dislocation + radial head + coronoid; fix coronoid, radial head (fix/replace), then LCL, MCL/hinged fixator last. LUCL insufficiency = PLRI; lateral pivot-shift test.**

Exam pearl: **Never excise the radial head with Essex-Lopresti, MCL or coronoid injury. Avoid overstuffing radial head arthroplasty. Pronate the forearm to protect the PIN in lateral approaches. Galeazzi = fracture of necessity; Bado II posterior Monteggia is commonest in adults.**

Fracture healing and fixation principles

1. Bone healing

Primary (direct) bone healing

- Occurs with **absolute stability** (anatomical reduction and **interfragmentary compression**, strain under about 2%).
- **Cutting cones**: osteoclasts at the tip cross the fracture, osteoblasts follow laying down lamellar bone (**Haversian remodelling**). **Contact healing** (gap under about 0.01 mm) and **gap healing** (gap under about 0.8-1 mm: woven bone first, then remodelling). **No callus.**

Secondary (indirect) bone healing

- Occurs with **relative stability** (casts, nails, bridging plates, external fixators; strain about 2-10%). **Callus formation** via endochondral (and intramembranous) ossification.
- **Stages:**
  - **Haematoma and inflammation** (hours to days): platelets, macrophages; cytokines **IL-1, IL-6, TNF-alpha, PDGF, TGF-beta**; mesenchymal stem cell recruitment.
  - **Soft callus (repair)** (about 2-3 weeks): chondrocytes produce **type II collagen** cartilage; periosteal intramembranous bone.
  - **Hard callus** (about 3-4 months): endochondral ossification, woven bone (**type I collagen**); **type X collagen** in hypertrophic chondrocytes during calcification.
  - **Remodelling** (months to years): woven to lamellar bone along stress lines (**Wolff's law**); **piezoelectric** effect (compression side negative: bone formation; tension side positive: resorption).
- **Perren's strain theory:** strain = change in gap length / original gap length. **Under 2%: primary bone; 2-10%: bone (secondary); 10-100%: cartilage/fibrous tissue; over 100%: granulation tissue/non-union.** A small gap with too much motion has very high strain (non-union); comminution distributes strain.
- **The "diamond concept"** (Giannoudis): osteogenic cells, osteoconductive scaffold, growth factors (osteoinduction), **mechanical stability**, plus

vascularity and host factors.

Growth factors and biologics

- **BMPs** (TGF-beta superfamily; bind serine-threonine kinase receptors, signal via **SMAD**). **rhBMP-2: FDA approved for open tibial fractures treated with IM nails** (BESTT trial) and anterior lumbar interbody fusion; **rhBMP-7 (OP-1)** humanitarian exemption for recalcitrant non-unions (withdrawn). Complications: heterotopic bone, swelling (cervical spine), possible cancer concerns.
- **Bone graft properties: osteogenic** (living cells: **autograft** only, and bone marrow aspirate), **osteoinductive** (BMPs; autograft, demineralised bone matrix), **osteoconductive** (scaffold: autograft, allograft, calcium phosphate/hydroxyapatite, tricalcium phosphate, calcium sulfate).
- **Autograft (iliac crest)** is the gold standard (all three properties). **Cancellous** autograft: rapid incorporation (osteogenic, creeping substitution); **cortical**: structural strength, slower (resorption first: weaker at 6 weeks-6 months).
- **RIA (reamer-irrigator-aspirator)** femoral graft: large volume, high osteogenic content.
- **Vascularised fibula:** for defects over about 6 cm, hypertrophies with loading.
- **Masquelet (induced membrane) technique** for segmental defects: **stage 1** debridement + **PMMA cement spacer** (induces a vascular membrane rich in growth factors, VEGF, BMP-2) ; **stage 2** (about 4-8 weeks) remove spacer, fill the membrane with cancellous autograft (± RIA).

Factors impairing healing

- **Smoking** (nicotine vasoconstriction, inhibits osteoblasts; **increases non-union** substantially), **diabetes**, **NSAIDs** (COX-2 inhibition; short courses probably safe; avoid in high-risk non-union cases), corticosteroids, **malnutrition** (albumin under 3.5 g/dL, lymphocytes under 1,500), vitamin D deficiency, infection, **excessive motion or excessive rigidity with a gap**, **open fractures with periosteal stripping**, high-energy injuries, **compromised blood supply** (scaphoid, talar neck, femoral neck, fifth metatarsal zone 2, tibial distal third), anaemia, hypothyroidism, quinolones (animal data), bisphosphonates (delay remodelling, do not stop union).
- **Pulsed electromagnetic fields, LIPUS** (low-intensity pulsed ultrasound: **TRUST trial** found no benefit in tibial fractures).

2. Non-union

- **Definition:** failure to heal with **no progression of healing for 3 consecutive months** (FDA: 9 months after injury). **Delayed union:** healing slower than expected.
- **Weber-Cech classification:**

Type	Biology	Appearance	Cause	Treatment
Hypertrophic ("elephant foot", "horse hoof")	**Biology good** (vascular)	Abundant callus that fails to bridge	**Instability (too much motion)**	**Improve stability**: exchange (reamed) nailing, compression plating; **no graft needed**
Oligotrophic	Moderate	Little callus	Inadequate reduction/distraction	Stability + biology
Atrophic	**Poor biology** (avascular)	No callus, rounded/pencilled ends	Poor blood supply, infection, smoking, stripping	**Debride to bleeding bone + bone graft/biologics + stable fixation**
Pseudarthrosis (synovial)		Fluid-filled false joint	Persistent motion	Excise + graft + fixation

- **Infected non-union:** debridement, deep cultures, dead space management, stabilisation, local and systemic antibiotics, staged reconstruction (Masquelet, Ilizarov bone transport).
- **Work-up of any non-union: exclude infection** (CRP, ESR, WBC, deep tissue samples, nuclear imaging), **metabolic/endocrine screen** (vitamin D, calcium, thyroid, PTH, testosterone, glucose; Brinker's endocrine abnormalities found in a majority of unexplained non-unions), CT to confirm.
- **Exchange nailing:** for **aseptic diaphyseal non-union** (especially femur): ream at least 1-2 mm more and insert a larger nail. Less successful in the tibia with bone loss and in metaphyseal non-unions.

- **Distraction osteogenesis / bone transport (Ilizarov): latency 5-7 days**, distraction **1 mm/day in 4 increments (0.25 mm every 6 hours)**, then consolidation (about 1 month per cm in adults; **healing index**). Too fast: fibrous tissue/poor regenerate; too slow: premature consolidation. **Corticotomy** preserves the medullary blood supply and periosteum (low-energy). Regenerate forms by **intramembranous ossification**.

3. Biomechanics of fixation

Absolute vs relative stability

	Absolute stability	Relative stability
Healing	**Primary (no callus)**	**Secondary (callus)**
Indicated for	**Articular fractures**, simple (two-part) diaphyseal fractures of the forearm, simple metaphyseal fractures	**Comminuted diaphyseal and metaphyseal fractures**
Methods	**Lag screws, compression plating, tension band**	**IM nails, bridging plates, external fixators, casts**
Requirement	**Anatomical reduction**	Restore length, alignment and rotation; preserve biology

Screws

- **Lag technique:** the thread engages only the far cortex (**overdrilled "gliding hole" in the near cortex**); compresses fragments; place **perpendicular to the fracture** (best compression) or bisecting the angle between perpendicular to the fracture and to the shaft.
- **Pull-out strength** depends on the **major (outer) diameter**, thread depth and pitch, and bone density. **Bending/fatigue strength** depends on the **core (root) diameter** (strength proportional to the **radius to the fourth power** for bending stiffness, third power for strength).
- **Position screws** (hold fragments without compression), **locking screws** (threaded heads lock into plates: fixed-angle construct), **cannulated screws** (smaller core: weaker).

Plates (function, not design)

Function	Description	Example
Compression	Dynamic compression holes (eccentric screw) or tensioning device compress a transverse fracture	Forearm shaft (LC-DCP)
Neutralisation (protection)	Protects a lag-screw fixation from bending/torsion	Lag screw + one-third tubular plate in a Weber B fibula
Buttress (antiglide)	Resists shear/axial load at a metaphysis	Tibial plateau, volar Barton's, posterior antiglide fibula
Tension band	On the **tension (convex) side** converts tension into compression	Olecranon wiring, patella, femoral plating laterally
Bridging	Spans comminution; relative stability	Comminuted shaft (MIPO)

- **LC-DCP** (limited contact): undercuts reduce periosteal contact, preserve blood supply, even stiffness.
- **Locking plates:** fixed-angle; do not depend on bone-plate friction; **advantageous in osteoporotic bone, short metaphyseal segments, periprosthetic fractures**; act as an "internal external fixator". **Too stiff** (short working length, all holes filled near the fracture) can **suppress callus** (**far cortical locking / dynamic locking screws** add flexibility). Bridging rule: plate span about **2-3 times the fracture length**, **screw density under 0.4-0.5**, leave 2-3 holes empty over the fracture.
- **Working length:** distance between the closest screws on either side of the fracture; longer working length = more flexible construct.

Intramedullary nails

- **Load-sharing** devices (vs load-bearing plates). **Bending stiffness proportional to the fourth power of the radius** (a slightly larger nail is much stiffer); slotted nails have reduced torsional stiffness.
- **Locking: static** (screws both ends: controls length and rotation), **dynamic** (allows compression; for stable patterns).
- **Reaming:** increases nail size and union rates; produces **autograft** at the fracture; temporarily reduces endosteal blood flow (periosteal flow increases

compensatorily); fat intravasation.

- **Poller (blocking) screws:** narrow the medullary canal in metaphyseal fractures to prevent angulation (placed on the **concave side of the deformity**).

External fixation

- **Stiffness increases with: larger pin diameter** (most important; stiffness proportional to radius^4), **pins closer to the fracture** and further apart on each fragment (**near-far spread**), **bars closer to the bone**, **more bars (double-stacked)**, **multiplanar** configuration, more pins.
- **Pin-site infection** is the commonest complication; **hydroxyapatite-coated pins** improve fixation in cancellous/osteoporotic bone.
- **Circular (Ilizarov) frames:** tensioned thin wires (90-130 kg tension); allow axial micromotion ("trampoline effect") that promotes healing; deformity correction and bone transport.

Material properties (implants vs bone)

- **Elastic (Young's) modulus**, from highest to lowest: **ceramic (alumina) > cobalt-chromium > stainless steel > titanium alloy > cortical bone > PMMA > polyethylene > cancellous bone > cartilage**.
- **Titanium:** modulus closest to bone (less stress shielding), excellent biocompatibility, **notch-sensitive**, poor wear resistance as a bearing.
- **Stress shielding:** stiff implants carry load, bone underneath resorbs (refracture after plate removal: protect for about 6 weeks).
- **Fatigue failure:** implants break if the fracture does not heal ("**race between healing and implant failure**").

4. Principles summary (AO)

1. **Fracture reduction and fixation to restore anatomical relationships.**
2. **Stability by fixation or splintage** appropriate to the fracture and injury (absolute vs relative).
3. **Preservation of the blood supply** to soft tissues and bone by gentle handling and careful reduction (MIPO, indirect reduction).
4. **Early and safe mobilisation** of the part and the patient.

Exam pearl: **Absolute stability = primary healing, no callus, anatomical reduction (articular, simple forearm shaft). Relative stability = callus (comminuted diaphyses: nails, bridging plates).** Perren: strain under 2% bone primary; over 10% fibrous.

Exam pearl: **Hypertrophic non-union = needs stability (no graft); atrophic = needs biology (graft) + stability. rhBMP-2 is FDA approved for open tibial fractures with IM nails.** Ilizarov: latency 5-7 days, 1 mm/day in 4 steps.

Exam pearl: **External fixator stiffness: pin diameter (most important), pins near the fracture and far apart, bar close to bone, double bar. Modulus: ceramic > CoCr > steel > titanium > cortical bone.**

Polytrauma, open fractures and compartment syndrome

1. Polytrauma: assessment and priorities

- **Trimodal death distribution (classic):** immediate (brain, heart, great vessels), early "**golden hour**" (haemorrhage, airway, tension pneumothorax: preventable), late (sepsis, multi-organ failure).
- **ATLS primary survey: ABCDE** (catastrophic haemorrhage first in military/C-ABCDE): Airway with cervical spine protection, Breathing, Circulation with haemorrhage control, Disability (GCS, pupils), Exposure.
- **Haemorrhagic shock classes (ATLS):**

Class	Blood loss	Heart rate	BP	Mental state	Base deficit
I	Under 15% (under 750 ml)	Normal	Normal	Slightly anxious	0 to -2
II	15-30%	Mildly raised (100-120)	Normal (narrowed pulse pressure)	Anxious	-2 to -6
III	30-40%	120-140	**Falls**	Confused	-6 to -10
IV	Over 40%	Over 140	Low	Lethargic	Below -10

- **Damage control resuscitation: permissive hypotension** (systolic about 80-90 mmHg, except head injury: MAP over 80), **balanced transfusion (packed cells : plasma : platelets about 1:1:1)**, **tranexamic acid (1 g over 10 min then 1 g over 8 h) within 3 hours** (CRASH-2), avoid crystalloid overload; correct the **lethal triad: hypothermia, acidosis, coagulopathy** (plus hypocalcaemia: "lethal diamond").
- **Injury Severity Score (ISS):** sum of the **squares of the highest AIS in the three most severely injured body regions** (range 1-75; AIS 6 = 75). **Polytrauma: ISS over 15-17** (or the **Berlin definition**: AIS 3 or more in two or more regions + one physiological factor: hypotension, GCS 8 or less, acidosis BE -6 or less, coagulopathy, age 70 or more).

2. Early total care vs damage control orthopaedics

- **Early total care (ETC):** definitive fixation of all long-bone fractures within 24 hours; reduces pulmonary complications and allows mobilisation in **stable** patients.
- **Damage control orthopaedics (DCO):** in unstable patients, the **"second hit"** of major surgery (reaming, blood loss) on top of the **"first hit"** (injury: SIRS) can cause **ARDS and multiple organ failure**. So: **temporary external fixation**, haemorrhage control, resuscitation in ICU; **definitive fixation once physiology recovers (window about days 4/5-10/14; avoid days 2-4** when the inflammatory response peaks).
- **Patient categories (Pape):**

Status	Criteria (examples)	Strategy
Stable	Haemodynamically stable, no ongoing transfusion, lactate under 2.5	ETC
Borderline	ISS over 40 (or over 20 with chest trauma), multiple long-bone + truncal injury, bilateral lung contusion, mean PAP over 24, hypothermia under 35 C	ETC with caution or DCO
Unstable	Persistent instability despite resuscitation, BP under 90, needing over 2-3 units/hour	**DCO**
In extremis	Uncontrolled bleeding, lethal triad	**DCO**: life-saving surgery only

- **Early appropriate care (EAC, Vallier):** definitive fixation within 36 hours once **lactate under 4 mmol/L, pH 7.25 or above, base excess -5.5 or above** (resuscitation endpoints), rather than fixed time rules.
- **Fat embolism syndrome:** 24-72 hours after long-bone/pelvic fractures; **Gurd and Wilson criteria** (major: **respiratory insufficiency, cerebral involvement, petechial rash** in axillae/conjunctivae/oral mucosa; minor: tachycardia, fever, retinal changes, jaundice, renal signs, anaemia, thrombocytopenia, raised ESR, fat macroglobulinaemia). Prevention: **early fracture stabilisation**; treatment supportive (oxygen, ventilation).

3. Open fractures

Gustilo-Anderson classification (graded **after debridement**)

Type	Wound	Soft tissue	Contamination / fracture
I	**Under 1 cm**	Minimal	Clean; simple fracture
II	**1-10 cm**	Moderate	Moderate contamination
IIIA	Usually over 10 cm	Extensive	**Adequate soft-tissue coverage of bone** despite injury; includes high-energy/segmental/comminuted and **farmyard or gunshot** injuries regardless of size
IIIB	Extensive	**Periosteal stripping, bone exposed: needs flap cover**	Heavy contamination
IIIC	Any	Any	**Arterial injury requiring repair**

- **Infection rates:** I about 0-2%, II about 2-7%, IIIA about 7%, IIIB about 10-50%, IIIC about 25-50%; amputation rates rise to about 40-50% in IIIC tibial fractures.
- **OTA Open Fracture Classification (skin, muscle, arterial, contamination, bone loss)** is more reproducible.

Management (BOAST 4 / BAPRAS-BOA standards; AAOS)

1. **Antibiotics as soon as possible, ideally within 1 hour** of injury (the most important factor in reducing infection). **Cefazolin (first-generation cephalosporin)** for all; add **gram-negative cover (gentamicin)** for type III; **penicillin (anaerobic/clostridial cover)** for farmyard/soil contamination (AAOS 2022 guidance: cefazolin for I/II, ceftriaxone or cefazolin + gentamicin for III). Continue until **24 hours after wound closure** (no longer than 72 hours total).
2. **Tetanus prophylaxis** (toxoid ± immunoglobulin by status).
3. **Photograph** the wound, cover with **saline-soaked dressing**, **realign and splint**, document neurovascular status. Do not irrigate or probe in the emergency department.

4. **Debridement (excision)** by senior surgeons: within **12 hours** for high-energy injuries, **24 hours** for others, **immediately** for gross contamination, vascular compromise or compartment syndrome. Extend the wound along fasciotomy lines, remove all dead tissue (**4 Cs for muscle: colour, consistency, contractility, capacity to bleed**), loose bone fragments without soft-tissue attachment.
5. **Irrigation: low-pressure, normal saline** (**FLOW trial**: **soap was worse than saline**; low pressure equal to high pressure, cheaper). Volume by Gustilo type (3, 6, 9 L traditionally).
6. **Stabilisation: external fixation** or **definitive internal fixation (intramedullary nail)** if the wound is clean and cover is planned. **Unreamed vs reamed nails: no difference (SPRINT)**.
7. **Soft-tissue cover: definitive cover within 72 hours** ("**fix and flap**" by a combined orthoplastic team), ideally at the same time as definitive fixation. **Negative-pressure wound therapy** as a temporary dressing only (**WOLLF trial**: no benefit over standard dressing for infection or function).
8. **Local antibiotic** (beads, coated nails) for type III injuries.

Limb salvage vs amputation

- **Absent plantar sensation is NOT an indication for amputation** (it was once taught as one). In the **LEAP study**, about half of initially insensate feet regained sensation by 2 years, and **salvage and amputation had similar functional outcomes (Sickness Impact Profile) at 2 and 7 years**.
- The decision is individual: **warm ischaemia over about 6 hours**, a non-reconstructable vascular injury, severe soft-tissue and bone loss, a poor host, and the patient's wishes. Secondary amputation remains an option if salvage fails.
- **MESS (Mangled Extremity Severity Score):** skeletal/soft-tissue energy, limb ischaemia (doubled if over 6 hours), shock, age; **7 or more** historically suggested amputation (**poor predictive value; not to be used alone**). Others: LSI, PSI, NISSSA, HFS-97.

Gunshot injuries

- **Low-velocity (handgun)**: often managed like low-grade open fractures (local wound care, oral/short IV antibiotics). **High-velocity (rifle)** or shotgun at close range: **cavitation**, extensive debridement (treat as type III). **Intra-**

articular bullets must be removed (lead synovitis/toxicity, mechanical damage). Bullets in the bowel then through bone: broad-spectrum antibiotics and debridement.

4. Vascular injury

- **Hard signs** (pulsatile bleeding, expanding haematoma, absent pulses, bruit/thrill, ischaemia): **immediate exploration/repair**. **Soft signs:** ABI.
- **Ankle-brachial index (ABPI) under 0.9 = arterial study (CT angiography)**; over 0.9 has a high negative predictive value.
- High-risk injuries: **knee dislocation** (popliteal artery; **ABI for every knee dislocation**), supracondylar humerus, Schatzker IV/VI plateau, distal femur.
- **Sequence:** temporary **vascular shunt** first (restores flow within the golden **6 hours**), then skeletal stabilisation, then definitive vascular repair, **prophylactic fasciotomy** after reperfusion (especially ischaemia over 4-6 hours).

5. Compartment syndrome

Pathophysiology and causes

- Raised pressure within a closed fascial compartment exceeds capillary perfusion pressure: venous congestion, muscle and nerve ischaemia (**muscle necrosis after about 4-6 hours, irreversible by 6-8 hours**; nerve function lost earlier).
- **Causes: tibial shaft fractures (commonest, about 2-10%; young men)**, forearm fractures, crush injury, **reperfusion after vascular repair**, tight casts/dressings, burns (eschar), prolonged lithotomy, drug overdose with limb compression, snake bite, bleeding disorders/anticoagulation, IV infiltration. **Open fractures do not exclude it.**
- Highest-risk groups: **young men** (muscle bulk), tibial and forearm fractures, high-energy injuries, **unconscious/ventilated/regional anaesthesia** patients (signs masked).

Diagnosis (clinical first)

- **Pain out of proportion** to the injury, **pain on passive stretch** of muscles in the compartment (earliest, most sensitive signs), **tense swelling**, then paraesthesia/hypoaesthesia (deep peroneal: first web space in anterior

compartment), weakness. **Pulses are usually present** until late (pulselessness is a late sign, or indicates arterial injury).

- **Compartment pressure measurement** (slit catheter/arterial line transducer, within 5 cm of the fracture, deep posterior and anterior compartments of the leg): **delta pressure (diastolic BP minus compartment pressure) under 30 mmHg** = fasciotomy (McQueen). Absolute pressure over about 30-45 mmHg was older criteria. **Continuous monitoring** in obtunded patients. Diastolic pressure under anaesthesia is lower: use pre-operative diastolic values.

Treatment: emergency fasciotomy

- Remove all circumferential dressings and casts to skin, place the limb at heart level (**not elevated**), correct hypotension, oxygen; then **fasciotomy without delay**.
- **Leg: four compartments** (anterior, lateral, superficial posterior, deep posterior):
  - **Two-incision technique: anterolateral incision** (about 2 cm anterior to the fibula; anterior and lateral compartments; **superficial peroneal nerve** at risk) and **posteromedial incision** (about 2 cm posterior to the medial tibial border; superficial and deep posterior compartments; **saphenous nerve and vein**). Release the deep posterior compartment by detaching soleus from the tibia.
  - Single-incision (lateral, perifibular) alternative.
- **Thigh:** three compartments (anterior, posterior, medial); lateral incision releases anterior and posterior.
- **Forearm: volar (Henry-type curvilinear) incision** from medial to the biceps tendon to the carpal tunnel (release **carpal tunnel and lacertus fibrosus**), ± dorsal incision for the mobile wad and dorsal compartment. **Deep flexors (FDP, FPL)** are most affected.
- **Hand:** 10 compartments (4 dorsal interossei, 3 palmar interossei, thenar, hypothenar, adductor pollicis): dorsal incisions over the 2nd and 4th metacarpals, thenar and hypothenar incisions; plus carpal tunnel release.
- **Foot:** 9 compartments (medial, lateral, 4 interossei, 3 central: superficial, adductor, calcaneal): **calcaneal fractures and crush injuries**; dorsal two-incision + medial incision.
- **Wounds left open**; re-look at 48-72 hours; delayed closure or skin graft.

- **Missed/late compartment syndrome (over about 24-48 hours, established necrosis):** fasciotomy is **not** recommended (high infection risk; converts closed necrosis to open infected necrosis); manage late contracture (**Volkmann ischaemic contracture**: flexor slide (**Max Page muscle slide**), tendon transfers, free functioning muscle transfer).
- **Complications:** rhabdomyolysis and **myoglobinuric renal failure**, hyperkalaemia, infection, contractures, sensory loss, chronic pain.

Chronic exertional compartment syndrome

- Athletes (runners); pain during exercise relieved by rest; **anterior and lateral leg compartments**. **Pressure criteria (Pedowitz): pre-exercise 15 mmHg or more, 1 minute after exercise 30 or more, 5 minutes after 20 or more.** Fasciotomy if conservative care fails.

6. Crush syndrome and rhabdomyolysis

- Prolonged compression (earthquakes, entrapment): on release, **hyperkalaemia (cardiac arrest)**, **myoglobinuria (acute kidney injury)**, hypovolaemia, metabolic acidosis, DIC.
- **Treatment: aggressive IV fluids before extrication if possible** (normal saline, aiming for urine output about 200-300 ml/hour), treat hyperkalaemia, consider bicarbonate/mannitol (debated), dialysis.

Exam pearl: **ISS = sum of squares of the three worst AIS regions (polytrauma over 15-17). DCO for unstable/borderline patients; definitive surgery after day 4-5, avoid days 2-4.** TXA within 3 hours.

Exam pearl: **Gustilo is graded after debridement; IIIB = flap needed; IIIC = arterial repair. Antibiotics within 1 hour; FLOW: saline, low pressure, no soap; definitive cover within 72 hours. LEAP: plantar insensation is not an indication for amputation.**

Exam pearl: **Compartment syndrome: pain on passive stretch is earliest; delta pressure (diastolic - compartment) under 30 mmHg = fasciotomy; pulses are usually present.** Leg: two-incision, four-compartment release.